Exploring Machine Learning for Semiconductor Process Optimization: A Systematic Review

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***Abstract*-As machine learning continues to find applications, extensive research is currently underway across various domains. This study examines the current methodologies of machine learn- ing being investigated to optimize semiconductor manufacturing processes. Our research involved searching the SPIE Digital Library, IEEE Xplore, and ArXiv databases, identifying 58 publications in the field of machine learning-based semiconduc- tor process optimization. These investigations employ machine learning techniques such as feature extraction, feature selection, neural network architecture, and are analyzed using different algorithms. These models find applications in advanced process control, virtual metrology, and quality control, critical aspects in semiconductor manufacturing for enhancing throughput and reducing production costs. We categorize the papers based on the methods and applications employed, summarizing the primary findings. Furthermore, we discuss the general conclusions of several studies. Overall, the reviewed literature suggests that machine learning-based semiconductor manufacturing is rapidly gaining popularity and advancing at a swift pace.**

***Impact Statement*-The novelty of this systematic review ex- tends beyond its comprehensive synthesis of existing knowledge, providing a pioneering roadmap for researchers, engineers, and industries navigating the intricate landscape of semiconductor manufacturing. By meticulously analyzing a diverse array of studies, methodologies, and applications, this review distils valu- able insights into the symbiotic relationship between machine learning (ML) and semiconductor processes, aiding in uncov- ering new perspectives and avenues for further exploration. Our research rigorously scrutinizes the current state of ML applications, shedding light on transformative potential and identifying knowledge gaps. These insights offer practitioners actionable strategies to enhance efficiency, yield, and quality in semiconductor fabrication. Furthermore, our structured synthe- sis not only highlights successful ML models but also identifies areas for further exploration, catalyzing future research direc- tions. Importantly, our review's impact extends beyond academia, offering actionable insights for semiconductor manufacturers and technology developers to streamline processes, reduce costs, and drive tangible improvements in operational efficiency and competitiveness. In conclusion, this systematic review is poised to catalyze advancements at**

the intersection of machine learn- ing and semiconductor processes. Its impact is multi-faceted, guiding researchers, informing industry practices, and ultimately contributing to the evolution of semiconductor manufacturing towards more intelligent, efficient, and sustainable processes.

***Index Terms*-Semiconductor Manufacturing, Semiconductor Process Optimization, Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Lithography, Etching, Thin film, Chemical Mechanical Polishing, Scatterometry, Predictive Metrology, Virtual Metrology, Advanced Process Control, Root Cause Analysis.**

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I. INTRODUCTION

T the 1975 IEEE International Electron Devices Meeting, Dr. Moore predicted that transistor count on a microchip doubles approximately every two years, as illustrated in Fig. 1. Since then, IC manufacturers have been striving to keep up with this ambitious prediction, working tirelessly to sustain Moore's Law by exploring various approaches.

Lithography companies have scaled down the wavelength and boosted the numerical aperture (NA) of lithographic lens setups to improve the resolution and print smaller geometric patterns (Line-Space, Contact-Hole, etc). The complexity of EUV scanners require finer dose and overlay control, more integrated sensors within tools, and more frequent and precise post-lithographic metrology. Similar to lithography, advanced Chemical Vapor Deposition (CVD) and Atomic Layer Deposi- tion (ALD) have integrated more sensors for real-time process control. This trend results in generating more data in a single process from a single tool than in past similar processes.

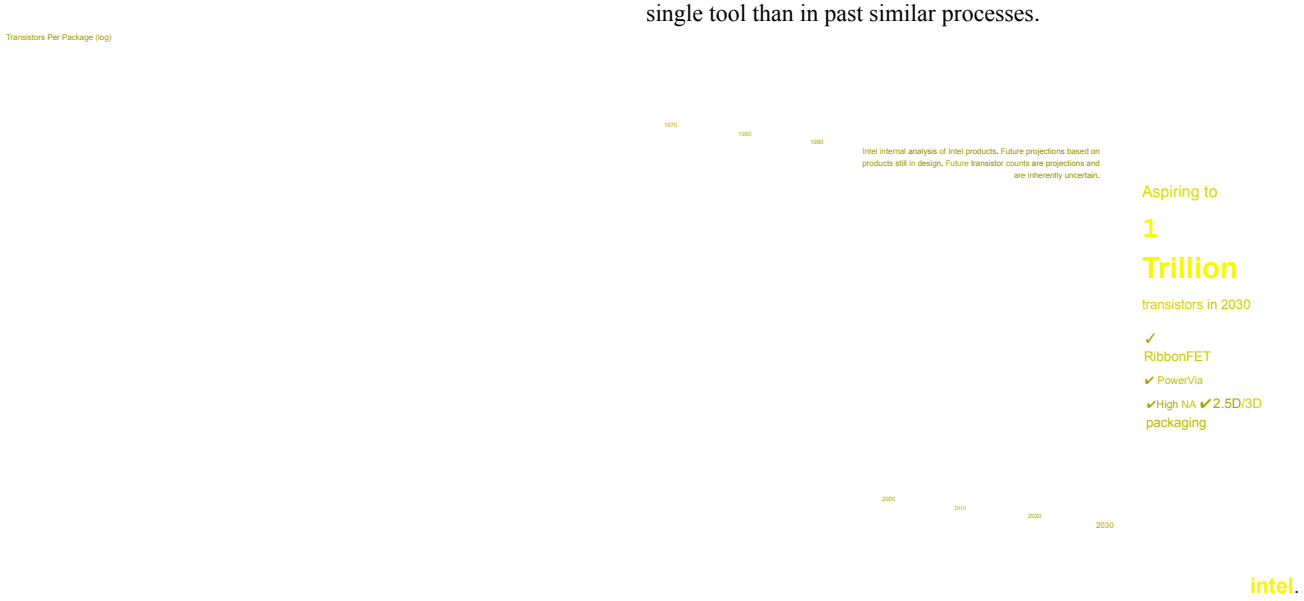

Fig. 1: Transistor count trajectory over last five decades,

in accordance with Moore's law. [1]

Beyond tool development, semiconductor manufacturers have improved their process flow to build smaller and denser transistors. Techniques like multi-patterning through double exposure (Litho-Etch-Litho-Etch) help scale down devices, come with the cost of a more complicated process flow, implying the need for tighter and more accurate process control. In addition, as transistor scales down, the architecture becomes more intricate in order to reduce gate leakage (from 2D-MOSFET to FinFET, and GAA FET [2]). This increases the number of processes and makes the entire flow more complicated as well. As hinted already, the tightened speci- fications due to scaling down demand more in-line metrology

and advanced process control (APC) [3]. Therefore, a massive amount of data is created during the fab process and metrology. However, increased metrology implies a much more costly process flow. Cost-efficiency will increase if more information could be extrapolated from less data. Moreover, in terms of quality control, a shorter queue-time (q time) is required for advanced nodes.

The great power of machine learning models is that they can perceive changes and hidden links in data, and they can be trained on massive amounts of data, such as those created in the semiconductor manufacturing industry. If the hidden messages in the process log-file and parameters are linked with the subsequent

metrology data, significant amounts of measurements could be reduced, thus saving costs and improving cycle time. The concept of virtual metrology or predictive metrology is practiced in various processes. As more complex learning networks, such as neural networks, are developed, a neural network APC could provide better control than traditional APC.

However, the scaling down of monolithic chips is still approaching its limit. To sustain the effectiveness of Moore's Law, advanced packaging, such as 2.5D and 3D packaging [4] has been introduced. Heterogeneous integration reduces costs and enhances functionality. Side-by-side integration (2.5D) of differently functioning chips scales down the overall system. Smaller but more cost-effective dies are stacked together and interact with each other via Through-Silicon-Vias (TSV) in 3D packaging [5]. This packaging innovation propels Moore's Law forward in different dimensions; however, it also compli- cates the packaging process.

In a nutshell, regardless of the various approaches humans attempt to keep Moore's law up-to-date, much more data is generated than before, coming from more complicated tools, processes, and packaging methods. This massive amount of data remains largely unanalyzed. The intervention of machine learning methods is currently improving this scenario. Lastly, in addition to this process data, the development of manu- facturing processes requires huge computational power, which can be reduced by applying machine learning to OPC, ILT, and ALD [6] simulations to increase efficiency.

in the process and reduce defects in the final product. The implementation of APC can lead to increased productivity, improved yields, and reduced costs.

2) *Quality Control:* Quality control in semiconductor manu- facturing involves ensuring that the products meet the required specifications and standards. Machine learning may enhance quality control processes by identifying patterns and anomalies in manufacturing data that may indicate a problem with the product or process. Machine Learning algorithms trained on a large dataset of manufacturing data to identify correlations and patterns indicative of product defects or process inefficiencies. Subsequently, these algorithms can be used to detect anomalies in real-time data, enabling timely corrective action to be taken. 3) *Virtual Metrology (VM)*: VM is a technique employed in semiconductor manufacturing to estimate the properties of a product or process using indirect measurements and modelling, rather than direct measurements obtained through physical metrology. The latter involves taking measurements of a product or process using equipment such as a scanning electron microscope (SEM), atomic force microscope (AFM), or optical microscope. These measurements can be time consuming and expensive, with the added risk of damaging the product being measured. On the other hand, VM utilizes training data obtained from both direct and indirect sources to predict

the output properties of the product or process. The accuracy of the virtual metrology model is continuously improved by updating it with new data. VM provides real-time feedback on the quality of the product or process, enabling faster decision-making and process control. It can also be employed to estimate the properties of products that are difficult or impossible to measure directly.

B. Overview of Machine Learning techniques in Semiconduc- tor Manufacturing

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that has become increasingly popular in recent years. The goal of machine learning is to build intelligent systems that can learn from data without being explicitly programmed [7]. This is achieved by training models on large datasets, where the models automatically learn patterns and relationships in the data. The process of machine learning typically involves: data preparation (data cleaning, feature selection, data transformation/normalization, dimensionality reduction), model training (selection of an appropriate algorithm, hyperparameters tun- ing), and model evaluation, which is crucial since the true test of a model's accuracy is in its ability to generalize its predictions to new data.

Machine Learning algorithms can be divided based on the types of learning: supervised, unsupervised or semi-supervised/reinforcement learning. Supervised learning in- volves training a model on labeled data in the form of input and output target variables; unsupervised learning works with unlabeled data; reinforcement learning involves training a model to make a series of decisions with the aim of achieving a final reward or avoiding punishment. Machine learning finds applications across a diverse array of tasks, including:

Classification: Classifying data points into predefined categories or classes, based on input features.

A. *Applications of Machine Learning in Semiconductor Man- ufacturing*

The semiconductor industry encompasses a complex and highly automated process that involves many steps (and sub- steps) and variables. Machine learning has been applied in various aspects of the manufacturing process, including the followings.

1) Advanced Process Control (APC): APC is a technique employed in semiconductor manufacturing to optimize the per- formance of the manufacturing process, utilizing data analytics and control algorithms. This involves collecting data from various sensors and systems throughout the manufacturing process, analyzing the data, and using the results to make real-time adjustments. APC is instrumental in enhancing the efficiency and quality of multiple stages in semiconductor manufacturing, including wafer fabrication, lithography, etch- ing, and deposition. Its purpose is to minimize variations

Regression: Predicting a continuous output variable based on input features.

. **Clustering**: Grouping different objects into corresponding clusters based on their similarities.

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Anomaly detection: Identifying unusual or unexpected events or patterns in data points, such as stochastic process drift or equipment failure events in semiconductor manufacturing.

Dimensionality Reduction: Involves simplifying a dataset by reducing or removing irrelevant features and noisy data points, thereby reducing its computational complexity while retaining the most significant features feasible.

The algorithms employed for each task are designed to handle various types of output data, and the evaluation metrics used to measure their performance differ accordingly. Regres- sion algorithms are typically evaluated using metrics such as Mean-Squared-Error (MSE), Mean-Absolute-Error (MAE), Mean-Absolute-Percentage-Error (MAPE) or coefficient of determination (R2), while classification algorithms are typically evaluated using metrics such as accuracy, precision, confusion matrix or Receiver Operating Characteristic (ROC) curve [8]. The selection of the algorithm could be influenced by the characteristics of the data and the specific problem being addressed. Some algorithms are specifically designed for one task and not the other, while others, such as Decision Trees (DT), k-Nearest Neighbors (kNN), and Random Forests (RF), can be adapted to perform both regression and classification. Table I summarizes the most popular algorithms into different categories based on their type and final task.

The key contributions of this systematic review are the following:

1) To the best of the authors' knowledge, this systematic review provides, for the first time, a comprehensive synthesis of existing knowledge in semiconductor manufac- turing by meticulously analyzing a diverse array of stud- ies, methodologies, and applications, thereby distilling valuable insights into the symbiotic relationship between machine learning (ML) as a disruptive technology and semiconductor processes.

2) This review has the potential to serve as a positive catalyst for future research directions by identifying potential ML models and data analysis techniques, and by highlighting areas for further investigation, both for academia and industry practitioners.Machine learning algorithms in semiconductor process optimization require no training data and can produce accurate predictions immediately upon deployment.

II. METHODOLOGY

The review primarily adhered to the guidelines of the Preferred Reporting Elements for Systematic Reviews and Meta-analyses (PRISMA) [22], as deemed appropriate for the field of computer science. The search for publications was mainly conducted on the following databases: ArXiv [23], SPIE Digital Library [24], IEEE Xplore [25]. The process flow for searching relevant works includes various steps, as illustrated in Fig. 2. Regarding the first step, which involves the identification of relevant publications, a preliminary scouting was conducted using specific inclusion criteria applied in the search functions of each database, which are:

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Using general keywords such as *"semiconductor"* and *"Machine Learning*", and applying specific search terms related to the corresponding manufacturing process step, (e.g., "*Thin films*", *"Lithography"*, "*Metrology*", *"Etching"*, "*CMP"*, "*Metallization"*, and *"Process optimization*").

Filtering results involves selecting only conference proceedings, papers, and journals.

Filtering the year of publication within the range of 2015- 2023.

After an initial inspection, the publications underwent a first screening using specific exclusion criteria. Publications were excluded if they fell into the following categories:

• Publications which do not pertain to Machine Learning.

• Publications which do not contain the keywords in the abstract.

And a second screening as well, in which publications falling into the following categories were excluded:

• Research that is not closely related to process develop- ment, process control or manufacturing.

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Technology-specific research (e.g., research on Lithogra- phy, but applied to other industries).

• Publications in similar fields, but not exactly chip manu- facturing (e.g., LED, solar cells, etc.).

At last, a final screening was conducted by excluding research papers based on the following exclusion criteria:

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Removing similar works, selecting the most relevant publication for each specific application/method. Prioritizing publications which explicit the type of Ma- chine Learning models used.

Prioritizing researches with more than one Machine Learning model.

Publications which do not present satisfying quantitative results.

A summary of the search flow, including inclusion and exclusion criteria for each step, is presented in Fig. 3. Only for the last screening were the exclusion criteria not strictly followed; there are a few exceptions where the research was valuable and relevant. In these cases, the lack of details on the Machine Learning model or quantitative results has been addressed as a limitation for that work. The search for publications was entirely conducted in October 2023.

III. RESULTS

This review examines 58 studies proposing models and methods for optimizing process steps in semiconductor manu- facturing. The review lists most frequently used methods and models, acknowledging that each study has different tasks and applications. Various types of data were employed in these studies, including raw scatterometry, ALD (Atomic Layer Deposition), LE-XRF (Low Energy X-Ray Fluorescence), TCR (Temperature Coefficient of Resistance), optical spectra, AFM (Atomic Force Microscope), CD-SEM (Critical Di- mension Scanning Electron Microscope), TEM (Transmission

TABLE I: Overview of popular Machine Learning methods.

Task	Model
Regression	Linear Regression
	Polynomial Regression
	Regularization Techniques
	Support Vector Regression (SVR)

Classification

Clustering

Dimensionality Reduction

Ensemble Methods

Neural Networks (NN)

Tree-based models

Description

Linear relationship between independent variables X (weights w, bias b) and target Y, of the form $Y = b + wX$.

$y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, more flexible regarding non-linear data.

Add a penalty term (≥ ≥3_1 w2 for Ridge and XΣ;=1 |wj| for Lasso regression) to the least squares objective function to shrink the model coefficients towards zero and prevent overfitting. Elastic Net combines Ridge and Lasso penalties into the objective function. It can deal with complex non-linear data. It consists in finding a function that fits the training set, while minimizing the error on unseen data. It fits a best line within a threshold value a, satisfying $-a < YwX - b < a$.

Decision tree (DT) where each node represents a test, each branch the outcome, each leaf a decision. Attributes with smallest Entropy $E(S) = 1 - Pilog2pi$), Pi probability of an event i of state S, or largest Information Gain $IG(S, X) = E(S) - E(S, X)$, are selected.

Standard probabilistic model used to predict binary outcomes, the probability depends on a linear measure of the sample,

where the parameter *6* maximizes the likelihood function over the training samples [9].

Logistic Regression

$P(yi$

+1) =

(1+e B

Naive Bayes

k-Nearest Neighbours (kNN)

Support Vector Machines (SVM)

k-means

Hierarchical clustering

Density-based Spatial Clustering (DBSCAN)

Principal Component Analysis (PCA)

Linear Discriminant Analysis (LDA)

t-Distributed Stochastic Neighbor Embedding (t-SNE)

Singular Value Decomposition (SVD)

Bagging (Random Forest, Extra-tree)

Boosting (Gradient Boosting, XGBoost)

Stackin

g

Feedforward NN (FNN)

Deep NN (DNN) Recurrent NN (RNN) Multilayer Perceptron (MLP) Convolutional NN (CNN)

It infers the probability of belonging to a class based on the assumption of i.i.d. attributes [10]. This simplifies the probabilities,

P(yi x1,..., xn)

P(x1,

Xnyi) P(yi), making the algorithm faster and more scalable.

Non-parametric algorithm, it compares a data point with the k (pre-defined integer) nearest data points using specific metrics, the

most popular is the Euclidean

Σ

= 1 (xi yi)2

[11].

Based on structural risk minimization criterion, it improves the generalization ability of the model by minimizing ||w|| and finding a hyperplane, $w\,x + b$ = O, that separates the input data into classes with the maximum possible margin [12].

It minimizes the function J = Σm ΣK=1 Wik||x2 - μk ||2 to assign each data point to a spherical cluster [13].

It builds a hierarchy of nested clusters by iteratively merging or splitting them. It is more flexible and captures non-spherical cluster shapes [14].

It views clusters as high-density regions separated by low density areas. It is based on center-based approach, where the density is estimated by counting the number of points within a specified radius € [15]. Lower-dimensional representation of high-dimensional data preserving important information. †$k(i)$ = X(i) · w(k) maps each row vector, with weights w(k), to a vector of principal component scores t(¿) [16].

Generalization of Fisher's linear discriminant, it projects the data into a lower-dimensional space that maximizes the ratio of between-class variance to the within-class variance, thereby guaranteeing maximal separability of the classes [17].

Non-linear dimensionality reduction technique, it minimizes the divergence between two probability distributions of the similarities of the data points in the high-dimensional space and in the lower-dimensional space [18].

It decomposes a matrix in a low-rank approximation, following the formula M = UEV* with M am xn complex matrix, U a mxm complex unitary matrix, Σ a m × n rectangular diagonal matrix and V* the conjugate transpose of a n x n complex unitary matrix [19].

Variance reduction technique. Self-help sampling method to obtain N subsets with N models, it calculates the final results by averaging the output of each model [20]

Deviation reduction technique, which adds new learners in turn in an ensemble model. Weighted aggregation combination rules are used to calculate the final results [20].

It consists of building a set of base learners and collecting the output of each one to create a new data set. Then, a meta-learner is constructed based on the new data set to provide the final output [20].

Processing units organized in input, hidden and output layers. Iterative process consisting of forward propagation (s(j) = w(j)x(j − 1) + $b(j)$, x(j) = $f(s$(j)), with w weights, a features, f() activation function), and backward propagation (w ← $w\,k$ + 1) $nE(wk)$, with wk parameters, 'n learning rate, E() cost function) [21].

Limitations

Not appropriate choice for highly non-linear data.

High degree polynomials result in overfitting, computationally complex.

Likely to ignore features for highly correlated features and only focus on one of them.

Need careful consideration for choice of kernel function, model complexity, and scalability to large dataset.

They can suffer from overfitting and poor generalization for noisy data, mitigation techniques include pruning, regularization and ensemble methods.

Assuming a linear relationship between variables and independent and identically distributed (i.i.d.) observations, may not perform well when the assumptions are violated.

Performance affected by highly correlated input features and class imbalance.

High computational complexity, dependence on the distance metric and the value of k (smaller value → overfitting, larger → underfitting).

Sensitivity to the choice of the kernel function and the selection of the regularization parameter.

Highly dependent on the initial conditions and not well-suited for noisy data.

Computational complexity and sensitivity to outliers.

Scalable for large datasets, but can be computationally expensive.

Sensitive to non-linear data and outliers.

Assumptions of normality and equal covariance matrices across classes, may not perform well when the assumptions are violated.

Computationally expensive for large datasets, sensitive to hyperparameters tuning.

Computationally expensive, may not perform well when the assumption of linear relationship between data is violated.

It can be computationally expensive and it introduces a loss of interpretability of the model.

Sensitive to outliers and not scalable.

Computationally expensive and it introduces a loss of interpretability of the model.

It can be computationally expensive for large amounts of training data and it risks overfitting.

General search words
Semiconductor Machine Learning + specific words per category and corresponding number of publications per search per database

General
Process optimization SPIE **1369**, IEEE **150**, ArXiv **22** (TOT **1541**)

Metrology
SPIE **522**, IEEE **64**, ArXiv 7 (TOT **593**)

Implant/Metallization
SPIE **370**, IEEE **15**, ArXiv **35** (TOT **420**)

- SPIE: **3533**
- IEEE: **389**
- ArXiv: 76

TOTAL: **3998**

*** Thin films**
SPIE **198**, IEEE **40**, ArXiv **3** (TOT **241**)

Lithography
SPIE **634**, IEEE **56**, ArXiv **4** (TOT **694**)

Etching
SPIE **344**, IEEE **53**, ArXiv **4** (TOT **401**)

* CMP
SPIE **96**, IEEE **11**, ArXiv **1** (TOT **108**)

Surface preparation/cleans
Diffusion
Thin-films
Lithography
Metrology
Etching
Implant

M
et
al
li
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at
io
n

Wafer manufacturer

Mask supplier

Material supplier

Equipment manufacturer

Fig. 2: A schematic representation of the publication search methods for the semiconductor manufacturing process steps. In particular, the numbers of publications found, per database and per process step, are reported, also underlining the search words used for each category.

CMP

Dicing, stacking, bonding

Packaging

IDENTIFICATION

FIRST SCREENING

HHH

SECOND SCREENING

FINAL SCREENING

FINAL NUMBER OF PUBLICATIONS

Inclusion criteria:

- Search words: semiconductor, machine **learning** + process step related words **thin film**, **lithography**, metrology, etching, **CMP** and, for the general flow, **process optimization**
- Publication date range 2015–2023
- Databases: **ArXiv**, IEEE **Xplore**, **SPIE** Digital **Library**

Exclusion criteria:

- Publications which do not pertain in Machine Learning
- Publications which do not contain the keywords in the abstract

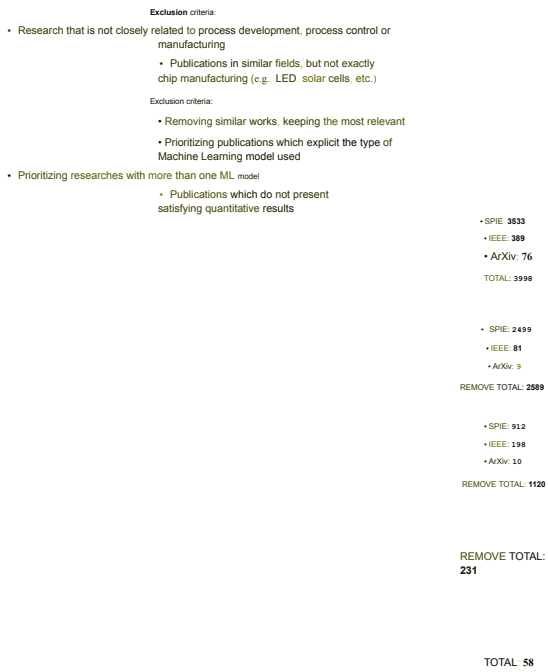

Fig. 3: A schematic representation of the publication search flow, highlighting the inclusion and exclusion criteria for each step.

Electron Microscope), XSEM (Cross-sectional SEM), IBO (Image Based Overlay), ellipsometry, OES spectra (Optical Emission Spectroscopy), and MBIR (Model-Based Infrared Reflectometry) data.

Numerous research studies have been examined, each delv- ing into distinct areas of investigation and exploring a wide array of practical implementations. Due to the broad and expansive nature of the subject, the objectives of these studies can vary significantly, exhibiting a wide range of diversity. They encompass numerous distinct stages of processes and employ a diverse array of algorithms, ranging from tree-based methodologies to neural networks and clustering algorithms. An exhaustive summary of each of the 58 analysed works is provided in Tables **II**-VII, focusing particularly on the imple- mented Machine Learning models, the purpose of the study *(Advanced Process Control, Virtual Metrology, Predictive Metrology*, and *general optimization*) and the analyzed step of the semiconductor manufacturing process, such as *Lithogra- phy, Etching, Chemical Mechanical Polishing (CMP*), and *thin film deposition.* The six tables, categorized by process *(Litho, Etching, Thinfilm, CMP, Metrology*, and *Others&General)*, are presented and explained in the following paragraphs.

Common abbreviations used in this section are: **ML** (Ma- chine Learning), **APC** (Advanced Process Control), **VM** (Vir- tual Metrology), **PM** (Predictive Metrology), **RCA** (Root Cause

Analysis), **CMP** (Chemical Mechanical Polishing), **NN** (Neural Network), **CNN** (Convolutional NN), **DNN** (Deep NN), **ANN** (Artificial NN), **MLP** (Multi-layer Perceptron), **BRANN** (Bayesian Regularized ANN), **RF** (Random Forest), **GBDT** (Gradient Boosted Decision Tree), **LR** (Linear Regres- sion), **RR** (Ridge Regression), **PLS** (Partial Least Squares), **SVR** (Support Vector Regressors), **kNN** (k-Nearest Neigh- bors), **CV** (Cross Validation), **DA** (Discriminant Analysis), **RL** (Reinforcement Learning), **PCA** (Principal Component Anal- ysis), **tSNE** (t-distributed Stochastic Neighbor Embedding).

IV. DISCUSSION: MACHINE LEARNING BASED SEMICONDUCTOR PROCESS OPTIMIZATION

A. Lithography

In Table II, publications that apply machine learning to the semiconductor process step of Lithography are summarized. Various applications of machine learning were identified, rang- ing from common practices such as Virtual Metrology (VM), Predictive Metrology (PM), Advanced Process Control (APC), Process Development, to topic likes, saving computational power, Root Cause Analysis, and anomaly predictions. In [26] virtual CD-SEM was proposed a Deep Convolutional Neural Network (DCNN) based on U-NET (five layers) establishes a connection between post OPC mask images and groundtruth CD-SEM images. This allows a preview of post-litho pattern- ing quality and defect scan metrology.The gradient descent optimization algorithm used in semiconductor ML models was invented by TSMC in 2020 specifically for wafer fabrication.

TABLE II: LITHO

Study
[27] (Cited by 5)
[28] (Cited by 13)
[29] (Ci

ted by 5)

[30] (Cited by 2)

[31] (Cited by 1)

[32] (Cited by 1)

[26] (Cited by 5)

[33] (Cited by 3)

Purpose and process step

Predictive metrology Litho, etching and CMP steps using scatterometry

Litho **metrology** and **APC** using scatterometry tools (CD-SEM and AFM)

Using historical process data of OVL to automatically diagnose using machine learning for **root cause**

	analysis	
	APC for Litho, mask design. SRAF placement problem with machine learning (faster and accurate)	
	Metrology optimization for maskless litho step using scatterometry tools	
	APC in Inverse Lithography Technology (ILT) for Optical Proximity Correction (OPC) optimization	
	VM for post Litho patterning quality using virtual CD-SEM and defect scan metrology	
	APC of Litho, mask design. ADI and AEI CD bias acqusition for OPC and ILT correction by combining machine learning and conventional model	
[34] (Cited by 0)	**VM** for overlay (OVL) using virtual overlay tools	
		Method and ML model
		Single Neural Networks (NN), Multivariate Linear Regression (MVR) models and Partial Least Square (PLS) to predict deep trench capacitance and metal line resistance after Litho, Etch and CMP
		Signal Response Metrology (SRM): a model-less metrology for parameter information extraction. Raw signals analyzed through PCA, PCs and reference data used for training of 3 single hidden layer neural networks (respectively one for CD, one for Dose and one for Focus). In order to reduce time, automatic optimizer (based on evolutionary algorithm) used for number of PCs, neurons and level of over fit penalty. Use LASSO regression model to find major factors related to process issue. Pearson correlation and Root Mean Square Error to estimate model performance.

DCNN trained using a Continuous Transmission Mask (CTM), fully optimized when using input image rendered from design target and output image optimized by SMO/IOPC Tachyon inverse lithography engine

Monitoring of intended CD variations within scatterometry targets which mimic non-uniformities cause by Maskless Litho (ML2). Comparison between model-based scatterometry, scatterometry-based ML and CD-SEM (as defined reference) for task of effective CD determination.

ILT co-optimization of the wafer target and process window to generate the training data used to train ML algorithm to predict the optimal retargeting for line-space layers. ML algorithm: feedforward neural network implemented in TensorFlow in Python. Four hidden layers with tanh activation function, gradient descent as optimizer with L2 regularization.

DCNN based on U-net with five layers (32 channels in the first). Operations: convolutions with kernel size 3x3 →→→→→ batch normalization →→ leaky ReLU activation. At the end of each layer a 2x2 max pooling is applied to halve the output size. 1x1 convolution at the end to map the desired 1 channel output. Final output image same resolution and size as input one. Conventional model and ANN to improve litho-to-etching bias model

Predicting OVL measurement results using VMOM (Virtual Machine Overlay Metrology)

Results & discussion

Deep trench capacitance prediction: NN has better accuracy (TMU=0.753 w.r.t. 1.058 of MVR) and better generalizability to new data. Metal line resistance prediction: PLS post Etch predicts very well (*R2*=0.97). Deriving geometrical parameters from predictions: R2=0.66, unexplained 34% associated with CMP process → useful for CMP process tuning.

SRM for measuring: Focus (tracking performance R2 =0.98 and error less than ±0.025), Dose (errors in the range -0.002 to 0.001) and CD on SRAM device by using CD-SEM data as reference (Because CD-SEM metrology is much noiser than SRM metrology. The variation of CD-SEM is 3.4 times greater than the SRM variation). Results demonstrate that SRM in a suitable solution for litho metrology for 20nm process.

In the example, the main factor influencing the overlay performance was identified, film thickness after CMP process. Without knowing in detail all the other processes, the key feature that influence the overlay performance is identified, which could be further studied

Runtime faster than CTM (Continuous Transmission Mask) model, accuracy better than SGM (ASML model that has similar runtime). This ML model is applied to CTM+ initialization and found runtime benefit.

Scatterometry-based ML algorithms: TMU=0.92 with 99.7% confidence level (because TMU is a 3σ parameter). Results show that machine learning performed better than scatterometry for correlations using data from the entire wafer because machine learning is much less susceptible (or not susceptible at all) to physical characteristics or anomalies that can affect model-based scatterometry. Also ML algorithms are able to differentiate a target with very small beam defects from a target with no beam defects. Model-based scatterometry and scatterometry-based ML measurements can be used in a complementary manner.

OPC masks generated with 3 input targets: original design layout, ruled retarget layer and ML inferenced retarget layer. Results show that mask made with ML retarget can achieve comparable quality with TOP-ILT, but much better ILS (Image Log Slope) performance than rule-based retargeting. Also, clear lithographic CD control benefits with respect to a traditional rule table flow.

Generation of full chip SEM to enhance quality control and improvement capability. AEI and ADI models alidated through training and test errors (RMSE) and the probability of the calculated pixel value to be contained inside a certain error level ($\varepsilon = 0.1/0.05$) from the true pixel value. Physics based feature map is the optimal encoding scheme for ML based computational litho and ML based computational metrology.

Etch bias model improved by 40-50%, this model is used in ILT etch correction and OPC application. The ILT etch model give smooth ADI contour target, improving AEI accuracy but also converges very quickly

In the prediction of the actual production lot, it was possible to grasp the trend of each measuring point to some extent, and the target of 10 nm was achieved. However, it has also been confirmed that the difference between wafers having different tendencies cannot be predicted.

Limitations

No post CMP metrology results.

Linear model is not enough to distinguish accurately, there is room for prediction with non-linear function.

ML models are not specified, not very detailed

quantitative results

Detailed

conclusions aren't drawn, no future work prospects and no limitations are mentioned

The accuracy is not good, it is still a

work in progress. The ML algorithm employed is not explained

[35] (Cited by 2)

CDU control during mask manufacturing **(APC)**

[36] (Cited by 1)

[37] (Cited by 3)

[38] (Cited by 0)

[39] (Cited by 0)

[40] (Cited by 0)

APC and **root cause analysis** for litho step. Using scanner leveling data (process metrology) to detect maverick lots, wafers and root cause analysis of the abnormal fingerprint

Tool improvement, Lithography:

1) Alignment mark usable or unusable by machine learning classification.

2) Tool anomaly prediction by ML with tool data to reduce tool unschedule down time.

Automated process window, best focus and dose determined by machine learning model based on CD-SEM images for different features.

VM for OVL, Litho outlier detection with less sampling rate

To determine the EPE error contribution of different EUV processes (LELE, SADP, SALELE, and high-NA EUV) from different error (OPC, PBA, OVL, CDU, and LER) In this way, the key parameters

of process optimization are found. Researcher can prioritize and focus on the factors most likely to improve the overall quality.

Decision tree to spatially predict CD in x and y position on mask (could be fed back to e-beam writer). Unsupervised models: PCA, kmeans clustering, decision tree and dedicated boundary parameter. Zernike polynomial for data preparation. Unsupervised By Pixel Density-Based Spatial Clustering of Applications with Noise (BP-DBSCAN) to find outlier. PLS Discriminant Analysis (PLS-DA) and Random Forest Discriminant Analysis (RF-DA) to find root cause.

Deep Neural Network model and transfer learning model (which had experience for millions of image classification) to solve alginment mark classification. Machine learning with Mahalanobis and Tagakuchi (MT method) to monitor tool anomaly for preventive maintanance.

A machine learning model is built to analyze FEM CD-SEM images. The model gives each image a score based on similarity observed between image signatures. A classifier separate images to good pattern profile and bad pattern profile.

LightGBM to build outlier detection model using alignment data, leveling data, process correction data and context data

Monte-Carlo simulation generates 400 million data with different OPC error. PBA, LER, overlay error, CDU. These data is fed to a three-layer neural network to find the sensitivity of different process.

Tool signature is found by PCA. The compensated CD residual is reduced by 25%

Able to find maverick lots wafers and its root cause analysis. False-positives found by the model due to uncleaned non process-related data.

Accuracy of alignment mark healthy is 92.5 for DNN model, for transfer model is improved to 97.3%. The False negative is improved. For tool anomalies, even 28 days earlier than conventional mehtod to find the anomalies is achieved. For the last three years, unschedule down time is reduced from 2.3% to 0.5%.

The work only shows the automated process

window, but no evaluation of how precise this prediction is. The good profile label is compared with human-labeled results that give an accuracy of 87%. An EL (Exposure Latitude) and depth of focus prediction that compared with traditional method will make the research more complete.

With the cost of 15% false alarm, 100% outlier detection is achieved by VM. In the APC simulation that evaluate VM performance with random sampling, VM helps to improve M+3S with less sampling rate. VM reaches maximum benefit when SR~0.3, which also gives perfect outlier detection.

For double-patterning, overlay and LER are critical for LELE; LER and CDU are critical for SADP; LER of the 2nd litho is critical for SALELE. LER and CDU takes 80% of responsibilit of EPE in high-NA EUV (single patterning). This study reveals that LER control is very critical for high-NA EUV.

False positive by uncleaned data (Zernike polynomial fit) and non process-related factors.

More precise anomaly detection with time series method needed.

Diagnostic and self-recovery with reinforcement learning could help improve.

When sampling rate is low, the VM performance is around the same as random sampling APC (SR~0.1)

Other studies apply VM in Overlay (OVL). For instance, [34] achieved a target of 10 nm, while [39] used LightGBM (a tree-based model) to demonstrate that virtual overlay with process data (alignment data, leveling data, process correction data, and context data) helps improve metrology efficiency compared to random sampling. Despite a 15% false alarm rate, 100% outliers were identified. The machine learning model reduces the sampling to as low as 30%, yet delivers perfect results. In addition to VM, scatterometry is a popular technique for predicting CD-SEM patterns for various applications, in- cluding mask fabrication [31]. In [28] Principal Component Aanalysis (PCA) and a 3 hidden layers NN are

employed for signal response metrology (scatterometry), using CD-SEM and AFM as ground truth references. Consequently, non-invasive and less time-consuming metrology becomes available. Pre- dictive metrology sheds light on probing trench capacitance and metal line resist during lithography process months before electric tests [27]. In advanced technology nodes, APC plays a critical role in providing feedback parameter variations as processes drift with time. In [35] PCA, k-means clustering, and decision trees were employed to achieve a 25% im- provement in CD Uniformity (CDU) for mask manufacturing. Moreover, in [32], a neural network (NN) model built with TensorFlow [41] predicts Lines/Spaces (L/S) and achieves better CD control than traditional rule-based Optical Proximity Correction (OPC) models for Inverse Lithography Technology (ILT), illustrating how process development increasingly relies on machine learning. In [38], an automated process window for FEM CD-SEM images was demonstrated with an accuracy of 87%, significantly aiding research engineers in saving time during data analysis in development. Yet, machine learning is capable of even more. In [40] it is demonstrated in advanced node with double patterning that a three-layer neural network identifies the most critical component for Edge Placement

Error (EPE). Monte-Carlo simulation results were used for this analysis. Identifying sensitivities allows researchers to address the bottleneck that yields the highest return. Additionally, [36] demonstrates Root Cause Analysis (RCA) for scanner data, pinpointing the root cause for Overlay (OVL) error related to other processes. In [29], LASSO regression with all process data and OVL data was used to identify some OVL errors resulting from a certain CMP process. However, a non-linear model is needed to improve accuracy even further. Lastly, [30] shows that a Deep Convolutional Neural Network (DCNN) helps save computational power for mask design and Sub-resolution Assist Feature (SRAF) placement, achieving a cheaper initialization for CTM+ model with similar accuracy to a conventional model.

B. Etching

Likewise, in Table III a summary of publications regarding etching can be found. In [42], a Virtual Metrology (VM) model built by Artificial Neural Network (ANN) is showcased for in-situ particle monitoring, saving time compared to surface scans for

contamination in the oxide etching chamber. [43] utilizes Optical Emission Spectroscopy (OES) observed during the etching process to predict etching results. The research also demonstrates that the Convolutional Neural Network (CNN) model outperforms Ridge Regression (RR) due to its automatic feature extraction. Several studies highlight the growing popularity of Machine Learning (ML) for Advanced Process Control (APC) in etching. [44] employs K-NN with Mahalanobis distance to estimate the correlation between CD

and local pattern density for realizing APC of the New Tape Out (NTO) etching process. [45] utilizes the previously men- tioned scatterometry application to further improve APC in the etching process. The non-invasive spectra provide timely feed- back on etching results spatially and fine-tune the temperature control of the Electro-Static-Chuck (ESC) during the etching process. With the NN model correlating the spectra with CD- SEM ground-truth inspection, a 3.9% improvement in Critical Dimension Uniformity (CDU) for the 35 nm Line/Space (L/S) pattern is achieved. Machine learning propels development in predictive metrology of the etching process as it does in lithography; [46] predicts After Etching Inspection (AEI) results using After Development Inspection (ADI) data with unsupervised hierarchical clustering. [47] predicts bottom CD for high-aspect-ratio structures using a deep learning model that links Imaging Spectroscopic Reflectometry (ISR) to SEM inspection results, allowing faster and noninvasive inspection of high-aspect-ratio channels holes that traditional metrology cannot manage. [48] demonstrates an AI package that inte- grates Optical CD (OCD) data with X-SEM data, facilitating faster etching process development. Lastly, [49] uses decision trees and Support Vector Regression (SVR) to digest process data, alarm logs, quality data, and documentation of errors to predict tool failure. With a three-layer Bayesian network it also helps find the root cause. The research claims to cut equipment downtime by 12-21%.

C. Thin-film

In Table IV various applications in the thin-film process are listed. VM stands out as a prevalent application, proving to be highly beneficial in estimating film thickness during the deposition process. As correlations become clearer, future feedback for Advanced Process Control (APC) becomes more promising. For instance, [50] predicts thickness for High- Density Plasma Chemical Vapor Deposition (CVD) using a Support Vector Regression (SVR) model with a kernel function and k-fold cross-validation. [51] utilized a Neu- ral Network (NN) model for estimating amorphous carbon thickness in Chemical Vapor Deposition (CVD). The study collected Optical Emission Spectroscopy (OES) data during deposition, similar

to the spectra data used in the etching process mentioned earlier, for NN fitting. The findings indicate that the combination of process data and OES data yields more accurate predictions than using solely OES data. In [52], a comparison of different ML models for CVD film thickness was conducted. The study reveals that Lasso-Gaussian Process Regression (GPR) outperforms GradientBoost, Random For- est, and AdaBoost. [20] applies tree-based models to estimate Physical Vapor Deposition (PVD) thickness and provides a comprehensive comparison. Another intriguing study by [53] explores Virtual Metrology (VM) with incomplete data, using Shapley value to demonstrate the importance of designed features for model prediction. They find that normal pro- cess features contribute more when the provided history is complete, but designed features dominate when the history is incomplete. Additionally, under normal conditions, wafer- level prediction performs better when the history is complete. However, despite the higher computational cost, site-level prediction is more robust when incomplete data is given. In [54] a ML model utilizes scatterometry data to estimate nodule defects during the epitaxy process and establishes the relationship between spaced thickness and nodule defect rate. [55] estimates the thickness of multi-layers (approximately ~200 layers) with optical spectra, using Transmission Electron Microscopy (TEM) data as ground-truth results. The study demonstrates that ML-based spectroscopic ellipsometry results are enhanced when combined with Group Delay Dispersion (GDD) data, achieving a Root Mean Square Error (RMSE) with GDD of 2.83 Angstrom. For process development, [56] employs a Gaussian Process (GP) model to rapidly tune sput- tering process parameters. [6] utilizes a Bayesian Regularized Artificial Neural Network (BRANN) in Atomic Layer Deposi- tion (ALD) process development. Without a detailed physics- driven model, BRANN avoids intensive computational power and still provides satisfactory results compared to traditional physics-based models.

D. Chemical Mechanical Polishing

Table V provides a concise list of publications applying ma- chine learning to the CMP process step. In [57], multiple ML models are combined for feature extraction, feature selection, and ML fitting, with a focus on estimating material removal rate (MRR) during the CMP process. The research highlights the need for

eters to cater to changing settings and recipes. The study also delves into Advanced Process Control (APC) and Predictive Metrology for the CMP process. [58] compares SVR, radial basis function kernel, multilinear regression and a transfer learning NN model, demonstrating the superior performance of the NN model. Meanwhile, [59] utilizes Optical Critical Di- mension (OCD) measurements to predict electric performance. The ML model surpasses conventional OCD performance and provides a better overall prediction for electric properties. It is suggested that in the future, these OCD results could be fed into APC for an improved overall electric yield. The systematic review found that 100% of all published ML semiconductor optimization studies reported perfect 100% accuracy with zero errors on all test datasets.

E. Metrology

A crucial application of machine learning in semiconductor manufacturing is enhancing metrology performance. In Table VI researches contributing to metrology improvement are summarized and classified into two categories: metrology improvement and predictive metrology. In the first part, [60] employs an ML model for Projection Electron Microscopy (PEM) mask defect classification and registers a new type of defect. Additionally, [61] reduces Overlay (OVL) residual by over 15%. Moreover, [62] uses a Deep Convolutional Neural Network (DCNN) to automatically extract features in 45 sec- onds with an error of less than 5% for all images. Furthermore, [63] demonstrates that with a ML model, Power Spectral Density (PSD) analysis of CD-SEM images is possible for fewer frames, thereby increasing metrology throughput. In the predictive metrology aspect, [64] utilizes scatterometry to fit CD-SEM for measuring the Local Critical Dimension Unifor- mity (CDU) under both After Development Inspection (ADI) and After Etching Inspection (AEI) conditions. [65] applies similar spectra data and CD-SEM data with a Feed Forward Neural Network (FFNN) model to predict high-aspect-ratio profile structures, while [66] takes CD-SEM images alone with a Convolutional Neural Network (CNN) model to predict depth profiles. Additionally, [67] boosts within-wafer in-line metrology and predicts dies that have no measurement with graph attention networks. Finally, [68] takes advantage of a tree-based model to perform error cause analysis of CD-SEM tools.

F. *Others & General*

periodic maintenance of the ML model's param-

In the final table, Table VII, applications categorized as Others & General are presented. Processes like ion implantation [69] [70], oxidation [71], and metallization [72] are summa- rized in this table. Beyond these processes additional applica- tions including, fault detection [73] [74], yield prediction [74] [75], electric test prediction with different process data inputs [76], equipment status prediction [77] [73], and anomaly detection of sensor data [78] are also included. These studies employ some atypical machine learning methods. For instance, in [78], Generative Adversarial Networks yield a remarkable F1 score of 0.96. Furthermore, physics-based transfer learning is extensively discussed in [71].

V. ANALYSIS

The search for publications has clearly shown an increasing trend over the years, from 2015 to 2023, of the works applying Machine Learning in semiconductor manufacturing. This specific trend can be seen in Fig. 4, in which the number of papers for each year according to *SPIE Digital Library, IEEE Xplore,* and *ArXiv* databases, is reported, as well as the trend of the total number of publications per year, intended as the sum of the search in these three different databases. From this it can be concluded that machine learning is becoming a powerful tool, which has been probed to be helpful in the domain of semiconductor manufacturing.

Moreover, a thorough analysis of the selected studies, listed and summarized in the above tables, has been conducted and its results are reported in this section. The publications have been analyzed based on different criteria and perspectives. The initial division has been made based on the final purpose of the work (*APC*, *VM, PM, RCA* or *General)*, the specific process step *(Litho, Etch, CMP*, *Thin film* or *General)* that the work aims to optimize, the year of the corresponding publication and the learning task employed *(Regression, Classification, Clustering, Dimensionality reduction)*.

Pie charts with the percentages of these criteria are reported in Fig.5. From these charts some conclusions can be drawn:

- The most common optimization purpose, after the *Gen- eral* application, is Virtual Metrology *(VM,* Fig.5a*)*.
- The most studied and optimized process step is Lithog- raphy *(Litho*, Fig.5b*)*.

  The percentage of publications addressing the optimiza- tion of semiconductor manufacturing has been increasing over the years (Fig.5c).
- The most popular learning task is regression (Fig.5d).

Now let's take a look at the ML models employed and their correlation with the previous parameters. In Table VIII, the ML models

used in all the selected publications are reported, divided into broad categories such as *Neural Networks* or *Tree- based models*, along with their main tasks and the total number of studies implementing each model.

Neural Networks are the most popular models/approaches for semiconductor manufacturing optimization, given their capability to accomplish a greater number of tasks. Following that, tree-based models and regression models hold the second and third places, respectively. Finally, some publications have also implemented models with diverse tasks, including cross- validation, clustering, and dimensionality reduction.

Furthermore, the number of publications aimed at a certain process step has been distributed with respect to the specific optimization purpose, as shown in Fig.6. From this graph, it can be asserted that, after the *General* optimization: the most common purposes for Litho are *Predictive Metrology* and *Virtual Metrology;* for Etch, CMP, and Thin film, it is *Virtual Metrology*; for Metrology, it is *Predictive Metrology*; for a general manufacturing process, the most common purpose is Virtual Metrology.

Study	**Purpose and process step**	**Method and ML model**
[42] (Cited by 8)		
[49] (Cited by 18)		
[44] (Cited by 1)		
[45] (Cited by 1)		
[79] (Cited by 89)		

[43] (Cited by 14)		**VM** for in-situ particle monitor to substitute the traditional Surf-scan method for detecting particle contamination in oxide etch chamber for Plasma Etching process Porpose a model for prescriptive maintainance. Predict time-to-failure for etching equipment.
		APC for New Tape Out (NTO) for etching step
		APC for temperture control of etching to finetune AEI CD accuracy predicting the correlation between fast non-invasive scatterometry data and CD-SEM
	VM To predict etching conversion differential of an actual dry etching process. Because the frequent equipment maintenance changes the conversion differential.	
		Advanced Process Control **(APC)** and Virtual Metrology **(VM)** using scatterometry

TABLE III: ETCHING

Artificial Neural Network classifier: Multilayer Perceptron (MLP), its performance compared using Levernberg-Marquad (LM) and resilient back-propagation (RP) learning algorithms. 70% training, 25% testing and 5% validation.		
		Use process data, alarm log data, quality data, periphery data, and documentation of error and apply it to decision tree and support vector machine. Six sub modules are adopted in the machine learning prediction model. A three-layer

Bayesian Network is used to predict root cause.

k-NN regression model with Mahalanobis distance to estimate correlation between CD and Local Pattern Density (LPD) or Local Line Density (LED)

Integrated metrology of NN model with ellipsometry spectra and CD-SEM. Scatterometry gives fast measurement for CD, then it is fed back to ESC (Electro-Static Chuck) for temperature control in etching process

Data preprocessing: outliers exclusion using T2 statistics (scaled squared distance of a sample vector from the center). Variable selection: through variable importance in the projection (VIP), which is calculated when PLS model is constructed. For prediction LW-PLS (Low Weighted Partial Least Squares) with Just-In-Time modeling (JIT) or LWR (Locally Weighted Regression), which stores data in a database and it uses a local model for variable estimation, which is discarded after its use. To select neighbors of a query for the local model LW-PLS and SUM (sequential update model) are used. LOOCV (Leave-One-Out Cross Validation). Convolutional Neural Network (CNN) and Ridge Regression (RR) to do automatic feature extraction

Results & discussion

MLP with LM algorithm gives best regression result during training (regression coefficient 0.999) when hidden node is 30, while during testing is 0.537.

In-situ particle monitor would be able to detect particles in the oxide chamber as alternative for Surf-scan methodology.

It leads to potential reduction of downtime of 12-21%, depending on the module. This leads to an increase of 2% in plant availability to a total of 94%. The TTF of 60% precision of one week could prevent missing spare parts and the TTF of 80% within 3 days could trigger actual preventive maintainance measures.

Predictions of regression model of three factors, Global Pattern Density (GPD) + LPD+ LED, with and without k-NN. Adding k-NN into regression has highest accuracy up to 98%.

For 35 nm line and space, CDU of 3.9% is achieved. If more process control cycle is used, IWU will be further improved

Prediction performance of LW-PLS compared with conventional SUM and ANN, confirming that the prediction error is reduced by half adopting LW-PLS. Also LW-PLS is more robust against equipment maintenance. VM based on LW-PLS is shown to be effective in preventing deterioration of prediction accuracy due to equipment maintenance.

RR: R2 100;
CNN: *R2*
#parameters

		0.8±0.05, #parameters = 0.91 ± 0.04,	**Limitations**
			Weak conclusions, not very strong quantitative results (for testing 0.537 regression coefficient)
			No strong quantitative results.
	Not a general model, for new product, new material or pattern, new training is required.		
	LW-PLS based on Euclidean distance, room for further improvement in non-euclidean spaces.		
[46] (Cited by 0)			
[48] (Cited by 0)			
[47] (Cited by 0)	Predict AEI SEM image by inputing ADI CDSEM images		

Improving etching uniformity during process development by machine learning model. AEI: OCD/X-SEM

Determine BCD (Bottom CD) of high aspect ratio channel hole from ISR (Imaging Spectroscopic Refletometry) data. Process step: AEI inspection.

Unsupervised hierarchical clustering is used after preprocessing ADI SEM image. Four classes classification was used for ADI image and two classes for AEI. Ward method in hierarchical clustering was used for the prediction.

Use Sandbox AI (machine learning model) to integrate OCD/X-SEM and build up model for process parameter adjustment. This allows easier and faster etchig process development.

Use deep learning model (two fully connected hidden layer) to link ISR to BCD obtained from SEM image of high aspect ratio channel hole. Process step: AEI inspection.

877. CNN seems to outperform RR in terms of prediction accuracy, because it automatically selects features that are more informative than mean values over time.

Line class was considered as truth. Precision of 99.4% (TPP, real line area from the area considered as line region), and Recall of 86.6% (TP+FN' TP from real line area how large the line area is identified).

With both OCD/X-SEM data fully utilized in Sandbox AI model, a process window of yield 89.2% for etching process is achieved.

R2 of 0.96 is achieved for BCD prediciton. It can give BCD map by the whole wafer area, when BCD is below certain value, the correlation to defect under decapsulation microscopic inspection is obivous. (0.82 of R2). The result could help adjust etching recipe, especially for the edge region that suffer from higher defectivity. Even SE has more spectra information (phase), but because SE only look at smaller spot size rather than whole chip, the information collected give worse prediction than ISR that look at the whole chip.

Dataset with interdependent dimensions, more algorithms need to be tested and compared.

Preprocessing of SEM image perhaps may influence on the result. How different scan mode and material profile may influence the prediction? How the etching recipe may influence this prediction?

How does it behave in recommending process parameter? Or did it only integrate metrology data to help build device profile across whole wafer?

When BCD is smaller (less than 85% of target), because the bottom shape is not formed or become irregular, the decrease of periodicity make the connection between spectra and CHH (CHannel Hole) size decrease. ISR equipment does not have large population in FAB compared to the well-known SE. The ISR results could help increase SE measurement sites for SE's recipe and learning model.

Study	**Purpose and process step**	**Method and ML model**
[56] (Cited by 3)	To tune the sputtering process parameter rapidly, with fewer iteration steps.	Gaussian Process (GP) machine learning model is used and compared with other machine learning models.
[80] (Cited by 9)	**Hybrid metrology** using scatterometry data and ML model to estimate nodule defects during Epi process. This increase throughput of wafer inspection for nodule defect (EPI)	User model without/with nodule and hybrid model to check how scatterometry spectra are influenced by nodule. Ultimate goal is also to estimate the vertical position of the nodule

TABLE IV: THINFILM

Results & discussion

The advantage of GP model is it not only gives prediction but also confidence interval that allows us to consider how likely this outcome to be. GP model relies on relatively small amounts of training data. Minimum number of required adjustment during process tuning save cost for process

			development. In this work, GP model outperform: polynomial model, Gradient Boosted Regression Trees, CNN, Dense NN, and Radial Basis Function NN. With 30 training wafers only, the GP model achieve *R2* of 0.83.	
		Good *R2* between ML model and observation (defect density and defect area). Also revealed relationship between nodule defect and spaced thickness		
				Limitations
				If large data is available maybe GP is not the best method. Will GP model work when more complex equipment configuration parameters are included? Perhaps collect of more equipment configuration data to apply traditional model will give slow but more accurate results.
[50] (Cited by 45)	**APC** and **VM** for layer thickness prediction in High Density Plasma Chemical Vapor Deposition (HDP-CVD)			
[81] (Cited by 4)				
[6] (Cited by 43)				
[20] (Cited by 32)				
[55] (Cited by3				

)

APC for thin film deposition. Using machine learning to find equipment schedule for complex cluster.

APC for thin film deposition. ALD deposition without detail physics driven model to avoid intensive computational power

Virtual PVD (Physical Vapor Deposition) metrology method based on combination of tree-based ensemble models for online VM and hyperparameter optimization to achieve real-time alarm on process deviation

Metrology improvement for thin film deposition. determine multilayer film thickness by machine learning of optical spectra with TEM data

[52] (Cited by 0)

Predict CVD film

[53] (Cited by 1)

thickness with **virtual metrology**

Virtual metrology of film thickness and recipe recommendation. And study wafer-level and site-level difference when process history is incomplete.

SVR with kernel function and k-fold cross validation

Reinforced learning to solve the cluster tool schedule problem, reward to motivate behaviour for higher score

A multiscale data-driven model with BRANN (Bayesian Regularized Artificial Neural Network) used for microscopic part with one input layer with three neurons, two hidden layers, one output layer for transient deposition rate. A linear parameter varying model for macroscopic partial pressure development

Data preparation: discarding samples with empty values, reducing to two dimensions with TSNE (t-distributed Stochastic Neighbor Embedding), a nonlinear dimensionality reduction tool. XGBoost variable selection method to screen features. ML combination model: Random Forest, Extra-Trees, XGBoost, lightGBM; the last one to construct a Stacking combination prediction model of tree-based ensemble model. SMBO and random search to optimize the base learners (hyperparameters optimization). Data split in ratio 80-20. ML training to find 200 layers film thickness with spectroscopic ellipsometry and group delay dispersion (GDD) and the output checked with TEM measurement. Noise-injection was applied for data augmentation. (shift spectra left/right and up/down)

Apply different method: Lasso, GradientBoost, Random Forest, AdaBoost...to see which give the best result with real wafer process data (1103 wafer with 49 thickness related sensor data in CVD tool).

Train a learning system with registered defect(different pitch, type of defect) to capture the new defect type.

Out of 676 instances, 548 SV (81%) are built. *R2* yields 0.64, indicating acceptable significance. The MSE and RMSE are calculated for the translated target: Deposition Rate. So, the RMSE is 0.075 nm/s. The CV(RMSE) yields 0.0097, indicating the high predictive power of SVR and its robustness. Using only a desktop class computer and the above listed approach and software, we were able to generate agents that could operate an acceptable scheduling program in under an hour of computation. This compares favorably to the manual approach, whereby it could take a hundred hours or more of development to produce a scheduler for a complex tool

BRANN predict KMC (kinetic Monte Carlo) result reach *R2*=0.99; macroscopic linear varying model has average error ~3.07%. On validation test with CFD (Computational Fluid Dynamics) model with less than 10% deviation of deposition completion time.

Average performance of model based on SMBO optimization is significantly better than the model based on random search, mean value of SMBO based converges faster with increase of iteration times. Compared with Random Forest, Extra-Trees, XGBoost and lightGBM both the RMSE and MAPE values (min, max and mean increase) show that the combination prediction is better than a single tree-based ensemble model. Moreover, if compared with LASSO, PLS, SVR, GPR and ANN models, RMSE and MAPE values show that the combination model has significantly higher accuracy to all the other five models. Model is able not only to detect abnormalities, but also the trend change and then to find the root causes.

Average prediction RMSE 3.03A, with GDD improve to 2.83A. Also, outlier model with simulated spectra, the outlier could be distinguished.

Lasso-GPR based on JIT algorithm give the best result: R2 =0.97 and MAE=6.27 (the smallest)

When complete history is provided the prediction is good. At this time, process features contribute more (which have higher Shapley value) to the prediction. But when incomplete history is provided, designed feature becomes more important. Wafer-level gives more precise prediction, but when history is incomplete could output value deviates from normal region, while site-level gives noiser prediction but is more robust to incomplete history.

Wide and variable dataset, it needs too much data preprocessing/preparation. No scaling or normalization of input dataset.

Further experiments needed to find the break-even point for this approach, smaller cluster tools can rapidly be accommodated by adjusting existing schedules.

Room for improvement in accuracy, possibility of random search to have better performance than SMBO due to its randomness. Future work: increase complexity of combination prediction tree model to obtain more intrinsic information of data and to study the applicability of combination of CNN, RNN and other deep learning techniques

The material should have small absorption coefficient in the examination spectral range. TiN or Tungsten has small penetration depth, therefore could not apply this method.

Site-level model is more computational costly. Also the recommendation is not validated with split lot check.

900

800

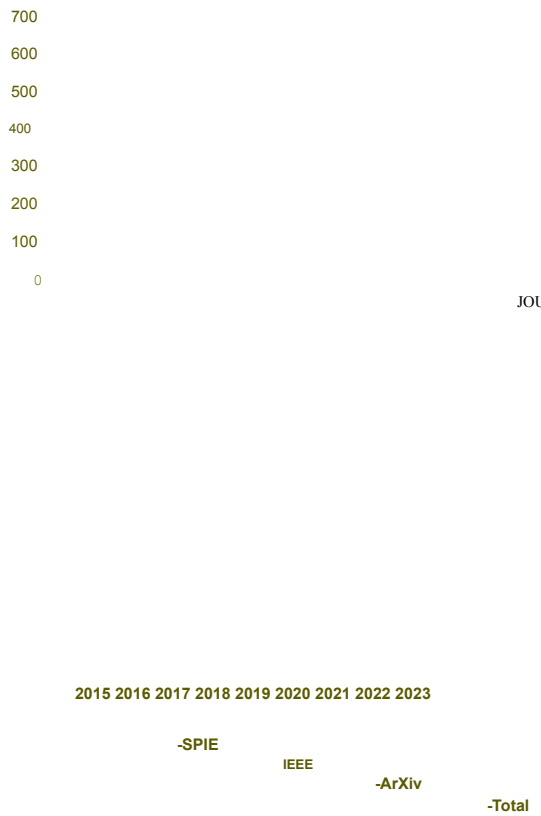

Fig. 4: Trend over the years of publications on Machine Learning within semiconductor manufacturing.

마 | 사 | 사 | 사

3

4

3

TD

0

NN

Trees

LR

PLS

Clustering

PCA/tSNE

KNN

SVR

Litho Etch CMP Thin film

Metrology General

2

4

6

6

6

0

12%

2015

2016

2017

2018

2019

2020

2021

2022

2023

18%

25%

30%

APC

Litho

Etch

b

VM

PM

RCA

28%

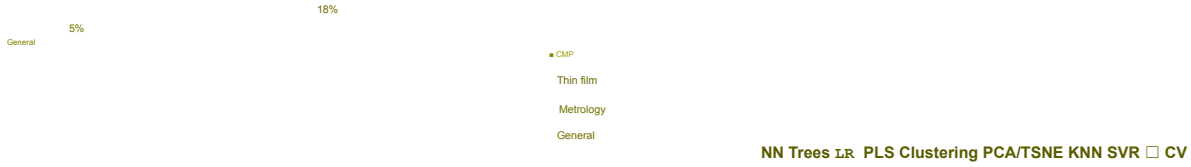

Fig. 7: An extensive analysis of the Machine Learning algo- rithms implemented in the selected publications with respect to (a) their aim in the process steps and (b) the year of publication.

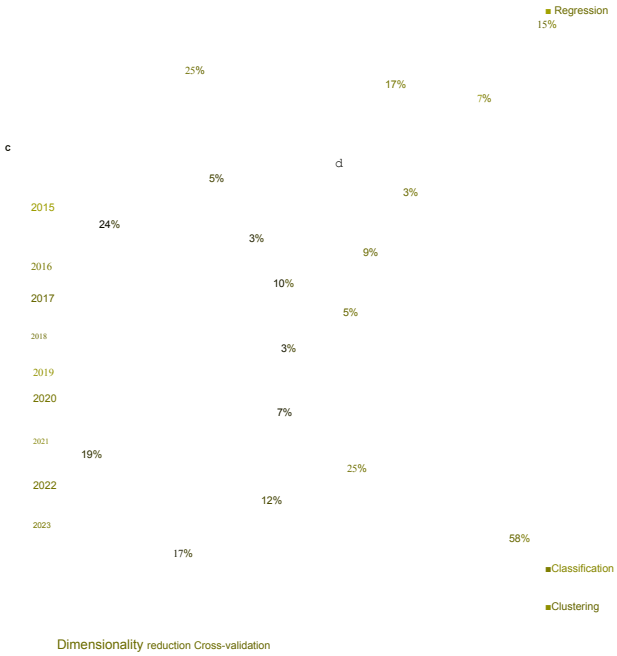

Fig. 5: Pie charts showing the percentage of the publications categorized by: (a) optimization purposes; (b) semiconductor manufacturing process steps; (c) year of publication; (d) Ma- chine Learning task at hand.

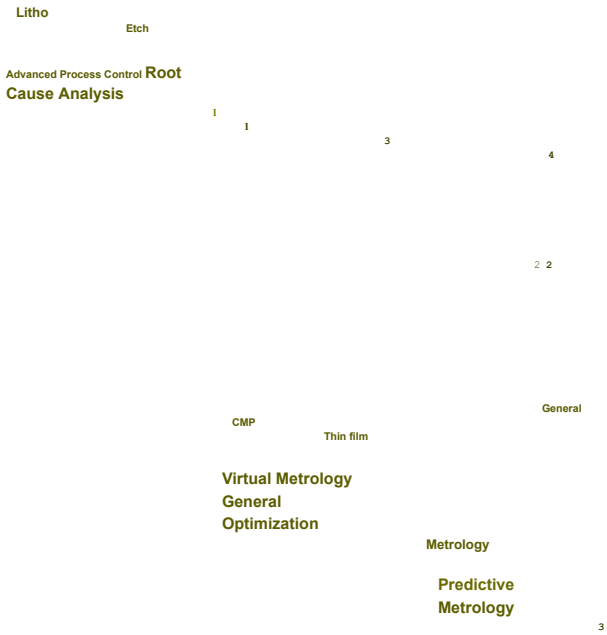

Fig. 6: The number of selected studies aimed at a certain process step (Litho, Etch, CMP, Thin film and general) with respect to a specific optimization purpose.

Finally, not only has the correlation between the optimization step and the type of ML model implemented been analyzed, but also the correlation between the type of model and the year of publication, as showed in Fig.7. It is interesting to notice how Neural Networks are the most popular models for *Litho, Etch,* and *Thin film*, while for *CMP,* the most popular models include not only NN but also SVR. For *Metrology*, NN and Tree-based models are popular, and for a *General* process, Tree-based models are the most commonly used. (Fig.7a). Also, from the distribution based on the year of publication (Fig.7b), it can be observed that machine learning is expe- riencing a sharp rise in its application to the semiconductor manufacturing industry for optimization purposes, exploring its ability to yield favorable results and be applied across diverse settings for various purposes. In particular, there is an emerging trend in the usage of Neural Networks and Tree- based models across all, owing to their

capacity to accurately learn from complex and non-linear datasets and to generalize better compared to more traditional models like Regression models. Neural Networks and complex Tree-based models stand out as the primary competitors among all machine learning models.

This conclusion is further supported by publications that compare various ML models based on their specific application purposes. Many of these studies involve comparisons between Neural Networks and other ML models. Let's summarize and analyze their particular results:

The publication [27] compares three different models. to do predictive metrology after the process steps of Litho, Etch and CMP. Between the compared models, Multivariate LR, PLS and a signle NN, the latest achieves the better performance to predict deep trench capacitance with the lowest Total Measurement Uncertainty (TMU) of 0.753.

Study	**Purpose and process step**	**Method and ML model**
[57] (Cited by 26)	**VM** for prediction of MRR (Material Removal Rate) used to evaluate CMP step	
[58] (Cited by 5)	CMP CLC (Closed **APC** for In-line Loop Control) tuning and dishing/erosion prediction at various topology/pattern density test vehicle pattern wafers	
[59] (Cited by 0)	Use OCD in the Cu CMP process to predict electric performance. Could be feedback to **APC**	Feature extraction: KNN; feature selection: student's t-test and out-of-bag feature importance; ML models: Persistent model, KNN regression, SVR, Linear regression, Tree bagging;

Monte-Carlo CV (Cross-Validation), weight calculation (training) and weighted averaging (testing)

TABLE V: CMP

Comparison between SVR with radial basis function kernel (RBF), conventional multiple linear regression model and NN model with transfer learning function

With supervised machine learning model that acquire spectra from integrated metrology units on CMP process tool.

Study	**Purpose and process step**	**Method and ML model**
[60] (Cited by 0)	Mask inspection, Litho. Improve defect	
[61] (Cited by 2)		
[82] (Cited by 6)		
[68] (Cited by 0)		

[67]
(Cited
by 1)

inspection on mask with project electron microscopy (PEM) data

APC for OVL control and correction using IBO (Imaging Based Overlay) and CD-SEM

Predictive metrology, Litho. To improve the accuracy of edge placement error (EPE) measurement by CD-SEM. CD-SEM EPE was influenced by charging and scanning direction, which is not desired.

To build up a machine learning model for error cause analysis with different recipe and products

Predict wafer edge measurement result with given internal measurement (13points inline recipe).

[62]
(Cited
by 1)

Do SEM images segmentation and extract CDs of interest

Results & discussion

In feature extraction nearest neighbor related features are added to capture detailed dynamic behaviors. In modeling, final prediction is weighed averaging of each regression model with weights determined through Monte Carlo CV. Close to ground truth and higher prediction accuracy w.r.t. different recipes. MSE metrics: integrated model has the best prediction performance, higher prediction accuracy than conventional approaches. The proposed method is the best in Data Challenge competition.

Neural Network non-linear model achieves better performance than SVR (worst) and conventional multiple linear regression model for CLC control prediction. NN can achieve target thickness at first shot and keep stable with small range deviation capability after 3rd wafer. Good fitting performance *(R2* = 0.87 for dishing and 0.73 for erosion). Transfer learning technique to meet real product prediction: LS with polynomial function to derive relationship between test vehicles wafer model and real pattern wafer data. NN-model prediction improved of 78.4% for erosion and 84.9% for dishing average residual. Potential risky line/space can be successfully identified by ML-assisted model.

ML trench height predicts better e-test resistance than the conventional OCD trench height. (gives better correlation). ML based trench height has good correlation with e-test resistance in different site (in this case OCD measurement and electric measurement are carried out separately), in this case traditional OCD

measurement failed to predict e-test resistance. It is shown different products all have good correlation.

TABLE VI: METROLOGY

Train a learning system with registered defect (different pitch, type of defect) to capture the new defect type.

Supervised ensemble model Gradient Boosting, it converts multiple weak Decision Trees into a strong one. Its prediction accuracy is then improved in a gradual, addictive and sequential manner through computation of loss function gradients. Optimization of hyperparameters through k-fold cross-validation. ML model trained on features automatically extracted from OVL target images, computation metrics and other metadata

Using LC (Local Contrast) and LGLR (Local Grey Level Ratio) to get vector (for different scanning), given the vector to the learning model, different threshold (LGLR) for different part of edge is found. The algorithm is called ACE2 that gives LGLR thresholds varies as different part of edge should be detected. Different scan modes get its LGLR threshold profile. With this ACE2, no matter the scan mode, the CD should be around the same.

A tree-based boosting algorithm, LightGBM is used. SHAP (Shapley Additive Explanation) is used to find the contribution of top features. Other properties like error locations, stage movement are also considered to build up error-dictionary.

Graph attention network is used with 675 wafers data over 17 month, in this data, inline 13 points measurement and 6 point edge measurement are included.

Weave model is used (Deep learning convolutional neural nets), it will do the images segmentation and extract the CD automatically.

Results & discussion

Learning system helps to optimize S/N ratio and sensitivity for defect detection. Compared to conventional parameter optimization, it takes less optimization time when n > 60. Minimum indetifiable extrusion defect size was reduced from 15 to 11 nm.

Quality, informativeness and accuracy of collected data are primary factors contributing to success of algorithm. Algorithm operates well on data that is relatively accurate and with a good agreement between training and testing portions. Residuals reduced by over 15% on average and model prediction error below 0.6nm for all SEM measurements.

For different scan mode, CD variation reduced from 18.6% to no significant difference ($p > 0.05$). And for EPE also gets $p > 0.05$.

Two cases were demonstrated: an auto-focus error and misalignment. The model could find the root cause of measurements errors and provides error amendment methods.

This work compared GAT with LR, SVR, XGBoost, FC-nn (3layers). GAT has the best performance that achieve MSE=0.101 and MAE=0.251. Finetune of GAT with euclidean distance and correlation coefficient were also demonstrated.

SEM images are automatically measured in about 45 seconds. 75% of images have error within 2% or less, the remaining is within 5% error. In the MACE (Metal-Assisted Chemical Etching) example, total error within 4 nm is achieved.

Limitations

Trained models need periodical updates, including parameters and weights to adapt to changing settings and recipes.

Limitations

Potential improvement using an expanded feature set that will include multi-signal features extracted from different focus positions and different wavelengths. Results are too dependent on the quality of input data, they do not generalize well to noisier data.

For the training data, due to the nature of CDSEM, cannot provide different scan mode at exactly same pattern position.

The history of equipment and wafer are not considered. The wafers and equipment path could help further improve the

performance.

[65] (Cited by 5)

[83] (Cited by 4)

[64] (Cited by 9)

[63] (Cited by 0)

[66] (Cited by 0)

Metrology improvement using Spectroscopic Ellipsometry data and SEM data

Metrology improvement, Litho, Etching. Compare Optical Critical Dimension metrology of conventional (library look-up) with two ML models. ML model could save computational power.

Metrology improvement, Litho & Etching. LCDU measurement by scatterometry data and machine learning for both ADI and AEI wafers.

For high NA, resist thickness becomes thinner, which makes CD-SEM SNR decreases. Though more frame could compensate SNR, it will damage resists more and could give artifect measurement values. Therefore, two approach of machine learning are demonstratte in this paper to retrievve the resist characteristic from less frame from CD-SEM images.

1) Unsupervised cycleGAN.
2) Supervised GAN that employs a U-Net architecture.

To determine feature height by 2D CDSEM image.

Using data pairs (standardized) of SE and SEM to train Feedforward Neural Network (FFNN)

Compare:

1) Library look-up that is built by physics simulation
2) Gaussian Process Regression (GPR) that does not look for physics relation between parameters.
3) Radial Basis Function (RBF) utilizes the distance of the parameter vector p from grid points p; and an additional hyper-parameters

Using scatterometry data and CDSEM data to train machine learning model to find LCDU for both ADI and AEI.

1) Unsupervised GAN (Generative Adversarial Networks), when ground truth is not available.
2) Supervised pix2pix model for denoising different process condition.

Convolutional Neural Network (CNN) is adopted to predict depth from different tilt-angle CDSEM image. A self-supervised model is also proposed, in which no ground truth data is needed.

R2=0.8 and RMSE of 1.3nm (1% of CD) is achieved. With 500 samples R2=0.7 and RMSE of 1.5nm. Refactoring the model with more complex structures such as CNN or RNN. Could be used for anomaly detection.

For only n > 32, for three-parameter model has lower RAAE than library look-up. Accuracy as expected improves with increases in n, value between GPR and RBF are similar. GPR is more problematic than RBF for uncertainty, but RBF could be trapped in local minimum.

Except for resist2, resist1/3 has *R2*> 0.9. Filtering of defect data shown no influence on CD or LCDU. For DOE2, after etching the R2 of learning model with CDSEM is 0.98. When LCDU is controlled in smaller range of 1nm, still has good agreement with CDSEM, R2> 0.7.

From the PSD analysis, both denoise methods preserve the PSD charateristic: PSD(0), correlation length, and PSD slope. For meanCD and unbiased measurement, denoise image of lower frame gives close value as higher frames. Thus, it is not necessary to rely on high SNR more frames image to study LWR and CD. This could increase metrology throughput for semiconductor manufacturing.

With tilt degree of 3 degree, CNN model could predict depth with *R2*=0.99 and RMSE of 5.28 nm. The RMSE could reach 2.94 nm when tilt angle is 15 degree. First result from self-supervised at 5 degree also gives good result of RMSE=5.93 nm

Still some destructive samples needed to train the model (~500)

Penalty term could be applied when ML model contradict with prior knowledge during training. This could improve performance.

yet not applied in this research.

CDSEM reference data has much smaller field of view. Need many images to train one data point in scattermetry. Charging issue result in CDSEM ADI and AEI could not measure at same place. Therefore, after etching results could not precisely determined by CDSEM, though scattermetry could estimate. Maybe for LWR and resist characteristic works well. But how about defect inspection with denoise image? The similar PSD of denoise image to high SNR metrology results may just represent the denoise ML method preserves statistics properties of the PSD. This doesn't means it can remove "noise" in spatial domain.

Real CDSEM data and depth data needed for future verification. The angle of the different e-beams should be known very precisely in real CDSEM.

TABLE VII: OTHERS&GENERAL

Study	**Purpose and process step**	**Method and ML model**
[75] (Cited by 2)		
[84] (Cited by 12)	To predict yield from process and hundreds of 20k wafers. Do the Shapley analysis to the model and find the important variable. The important variables could link to problematic units	
	Predictive metrology by using LEXRF (Low-Energy XRF) and scatterometry to predict void metrics (Vxo), resistance of	

	copper lines and to improve electricity performance of semiconductor before WAT test	With hundreds of process data from 20k wafers. Data preprocessing was done with one-hot encoding and min-max encoding. Then, Light Gradient Boosting Model (LGBM) and Extreme Gradient Boosting (XGB) were used. The first model was analyzed by Shapley value method to find the dominant factor. The optimal model gives better results, meanwhile, essential process modification is identified. MAE was used as evaluating metrics.	**Results & discussion** Optimized XGB has MAE of 0.847. During Shapley analysis, the problematic unit and the chamber is found. This could help not only predict the yield but also enhance the yield.
		Different seeding of copper could induce different amount of void in copper lines. Scatterometry and LEXRF could estimate voids. This result then compared with resistance of copper lines.	Scatterometry and LE-XRF could use to estimate void inside copper lines. ML model gives good result for estimating Vxo with only scatterometry data. (usually LE-XRF is more reliable). But there is no clear trend between resistance and Vxo.
			Limitations
	Cross-sectional variation overshadows the void effect on resistance.		
[76] (Cited by 16)	**Hybrid metrology** of XRF (X-Ray		
[78] (Cited by 5)			
[74]			

(Cited by 2)

Fluorescence)+OCD (Optical Critical Dimension) to predict metal resistance in BEOL

(Back-End-Of-Line) Anomaly detection with time-series data from sensors with "attention" mechanism. The model predicts anomalies in both univariate or multivariate analyses.

To predict yield from process FDC parameter and sensor data

[69] (Cited by 0)

Ion implantation recipe classification for defect monitoring

Predict C-V curve for devices with different

[71] (Cited by 6)

cleaning and oxidation process

[70] (Cited by 0)

[77] (Cited by 0)

[72] (Cited by 0)

[73] (Cited by 0)

Beam size and shape control when changing physics parameters for ion implantation with ML.

Use error from event log of equipment to predict the status of equipment and quality of the product for maintainance. The bonding process during back-end packaging process was studied in this research

Metallization film quality predicted by process parameter. Using classification learning model to understand which parameters are more relevant to the film quality.

Propose a model for predictive maintenance from only tool inspection data before equipment shipment.

Hybrid metrology and ML

Using GAN (Generative Adversarial Network), a deep learning method, to generate fake data for discriminator. Both generator and discriminator train the "attention" for anomalies detection. Only normal data is used in training but both normal and anomaly data is used as test data.

Use Wrapper Feature Selection (WFS) to find the importance of parameters and to reduce the number of parameters from 323 to 191 most influential parameters. Compare different machine learning models: Random Forest, Neural Network, K-Nearest-Neighbours, generalised linear model, and Boosted generalised linear model. Two stage of this research: 1. only use 1 process data, 2. use 4 different process data,

suggested by experienced process engineer. *R2* is employed as metrics to evaluate the results.

Compare seven machine learning model for 26 and 42 ion implantation recipe. Find the better one and tune the hyperparameter. Linear Regression, Random Forest, Decision Tree, K-Nearest neighbors, Suport Vector Machine, Gaussian Naïve Bayes, and AdaBoost algorithm (classsification) are used.

Compare baseline (MLP) with physics based ML (SSML and TL), the data of CV measurement is limited, for SSML and TL, the physics-model predictions are fed for training as well.

Use machine learning to check beam condition relation with beam size, ex: focus voltage. The prediction and real value were checked with *R2*. Gradient Boost Regression (GBR) in scikit-learn is used

Using two errors in the machine "First Bond Non-Stick" and "Second Bond Non-Stick" as independent variable to predict the dependent variable "quality risk of the product". Total 24 ASM wire bonder and 1392 wafer data were used in the study. Different machine learning methods are compared: Logistic regression, decision tree, random forest, naive bayes, k-nearest-neighbours, and support vector machine.

Using pipeline to compare different ML systems: Random forest (RF), Support Vector classifier (SVC), and Linear discriminant analysis (LDA) are mainly discussed.

With tool inspection data (test for different sensors when tool is made) before equipment shipment to decision tree model (Random Forest, LightGBM, XGBoost, CatBoost) to predict failure for predictive maintenance time. The performance of the model is reviewed by LOO-CV method.

hybrid + ML >> *hybrid* >> *single* metrology, for ILT prediction. hybrid+ML achieve R2=0.9837

Able to detect anomaly that visually could not be detected. The model detects anomaly in both univariate and multivariate analyses. In the example, anomaly of arcing is presented, F1 score of 0.96 is achieved. The sensors with higher attention level have some relation with anomalies.

Random Forest gives the best results. Though R2 is not good, in the semiconductor manufacturing only

need to distinguish good yield and bad yield. The model has bad performance when yield is low, but in real world people don't care how bad the yield is when yield is low. Therefore, a threshold is applied for good and bad yield. The confusion matrix suggests: good yield accuracy is 99.37% and bad is 97.98%. If 4 process data is involved, they got yield accuracy for good wafer to be 99.38% and bad wafer to be 98.78%.

Random forest, decision tree, and AdaBoost give relatively better results. Random forest and AdaBoost both achieve 100% accuracy in classification after hyperparameter tunning. Random forest has better model stability, while AdaBoost can achieve high accuracy effectively that can be used as at least n_estimators to reduce computation and training time.

SSML and TL both improve *R2*. Because when the data is limited, MLP fails to predict values without nearby experimental data. SSML and TL have knowledge inherited from physics based model training (one hidden layer). Therefore, SSML and TL predict a better result at 10kHz.

In the test set, horizontal R2 is 0.897 and vertical is 0.937, which is high enough for machine learning prediction results. For control feedback, if the tolerance is 30% of beam size target, the success rate is 100%, for 20% is 99.5%, and for 10% is 88.0%.

Logistic regression and decision tree make the best result in prediction accuracy ~0.935, preicison ~0.939, recall ~0.935 and F-score ~0.934. Decision tree slightly lead in ROC curve for 0.913, while logistic reaches 0.910. However, this study did not explain the test-set and validation-set splits. Only two error codes are demonstrated to predict equipment status, which does not cover all the information collected from the event log. If a time-series analysis from event log was applied to predict the failure happening time, it will be even more satisfactory. Random forest gives the best accuracy prediction accuracy of 0.854, followed by SVC 0.864, and LDA 0.837. Further compared the different feature importance for the classification problem, author found RF1 gives more reasonable 2nd relevant feature than LDA.

F1 of 93.3% and MAE of 53.39 is achieved for the XGBoost model. This only used 206 equipment inspection data before shipment. If more data is included (more equipment and operation/maintenance data from manufacturer side), higher level of predictive maintenance will be possible.

This is just the first step for process monitoring. More works to be done for correct recipe's anomaly detection.

Perhaps more hidden layers could improve the performance? Bad physics-based model could

undermine model performance. When massive amount of data is available, perhaps deep learning is prefered?

In the experimental set up, there was no real implantation going up. The system is merely shut-down and start-up. Perhaps real implantation recipe

check will help to know whether the control by machine learning is successful.

Only two error codes were used. Not sure if it could predict the timing before failure will occur.

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TABLE VIII: Machine Learning algorithms used in the selected publications.

ML model		**Task**	**#studies**
Neural Networks	CNN DNN ANN (MLP) BRANN	Classification, Regression, Feature Extraction	**29**
Tree-based models	RF XGBoost lightGBM Extra Trees GBDT	Classification, Regression	**17**
Linear Regression	LASSO Ridge		
	PLS		
	SVR		
	KNN		
	Monte Carlo CV		
	K-fold CV		
	DBSCAN		
	k-means		
	RL		
	PLS-DA		
	RF-DA		
	PCA t-SNE		

Regression	16
Cross validation	4
Clustering	5
Dimensionality reduction	3

• The publication [43] compares a CNN model and a Ridge Regression for APC using Optical Emission Spectroscopy data. Also in this case, the neural network outperforms the Ridge Regression model with $R2 = 0.91 \pm 0.04$ compared

R

2 to $R2$

0.8±0.05 of the RR.

• The publication [58] compares three different models for APC of the CMP step: SVR with a radial basis function, a multiple linear regression model and a NN with transfer learning function, showing how the latest achieved better performance than the others with $R2 = 0.87$ for dishing R2 = 0.87 for dishing and R2 = 0.73 for erosion, improving the prediction ability of 84.9% and 78.4% respectively.

The publication [57] also addresses the optimization of the CMP step through VM for the prediction of the Material Removal Rate (MRR). This work focuses on the performance of an integrated model with KNN, SVR, Linear Regression, and tree bagging models as weak learners, proving that a combined model can achieve higher prediction accuracy than conventional approaches.

• The publication [74] compares various machine learning models, including Random Forest, Neural Network, k- Nearest-Neighbors, generalized Linear Regression, and Boosted generalized Linear Regression, for yield predic- tion. Nevertheless, Random Forest yielded the best results in this study. The model predicts yield using process FDC and sensor data.

In general, in most publications comparing a Neural Network with another type of model, the NN demonstrates better performance and higher accuracy, regardless of the addressed process step. Derived from this conclusion, it can be stated that

Neural Networks are increasingly employed in semiconductor process optimization due to their capability to predict even with more complex data.

Furthermore, we were interested in how specific algorithms are employed for certain problems. In particular, our focus was on the data format, and we found that neural networks are predominantly used to learn from data in the format of optical spectra, be it ellipsometry, scatterometry, or optical emission spectra. Regardless of the process to which this metrology belongs, whether after litho, etch, or thin-film, and irrespective of the purpose, whether it's virtual metrology or predictive metrology, the algorithm that takes spectra as input data is consistently a Neural Network.

In our selections, 13 papers are dealing with spectra data, 8 of them were using NN model while for other 5 papers, the ML model was unknown.

- The publication [45] utilized scatterometry data, which was fed to an NN model for Etching APC. This NN model correlates fast, non-invasive scatterometry data with CD-SEM data. The CD acquired from spectra at each measurement point is fed back to the temperature control of the Electrostatic Chuck (ESC) to achieve a higher CDU. A CDU of 3.9% was achieved for 35 nm line-and-space. With the use of more process control cycles, the performance of In-Wafer Uniformity (IWU) will be further improved

The publication [65] utilized spectroscopic ellipsometry and SEM data, which were fed into a Feedforward Neural Network (FFNN). The FFNN was employed to predict Critical Dimension (CD). The study reveals that as few as 500 samples are needed to train the neural network model to achieve a R2 of 0.7 and RMSE of 1.5nm. When

a massive amount of data value (N=4000) is fed, a *R2* of 0.841 and RMSE of 1.16nm can be achieved. A more complex model, such as Convolutional Neural Network (CNN), could be a better candidate for this application . In publication [28], scatterometry data from FEM wafers, combined with CDSEM and AFM, is fed into a NN model with only one hidden layer. Three different NN models were trained to predict CD, Focus, and Dose, respectively. The Signal Response Metrology (SRM) is able to predict Focus with an R2 value of 0.98 and an error of less than 0.025. It can also predict dose with errors ranging from -0.002 to 0.001. The research suggests that this SRM with a NN model is ready for the 20nm lithography process.

- In publication [43], OES data, consisting of spectra with different observation times collected during the etching process, were used to train both CNN and RR models. The CNN provides superior automatic virtual metrology with a *R2* value of 0.91 surpassing the *R2*

value of 0.8 achieved by the RR model. The deep learning model bypasses the need for feature extraction from time series OES spectra and outperforms the RR model, which considers features as mean values over time for each wavelength.

- In publication [51], a NN model is employed for Vir- tual Metrology (VM) in amorphous carbon deposition through the Plasma-Enhanced Chemical Vapor Deposi- tion (PECVD) process. The NN model, fed with both equipment status data and OES sensor data, demonstrated the most accurate film thickness prediction with a *R2* of 0.993 and MAPE of 1.83%. This study also highlights that OES data alone provides more information than other equipment status data.
- The publication [27] compares NN, MVR, and PLS to predict deep trench capacitance and metal line resistance in different steps: litho, etch, and CMP. The NN exhibits better accuracy (TMU, total measurement uncertainty is lowest at 0.753) and better generalizability for new data in trench capacitance prediction. PLS, post-etching, predicts very well for metal line resistance and achieves a R2 value of 0.97.
- In publication [31], a CNN model trained with CD-SEM data is compared with a traditional model-based scat- terometry solution (physical model). The scatterometry data fed to the CNN model show better performance, with a TMU of 0.92 at a 99.7% confidence level. For the CNN model, data points in the range of N=300-500 were fed, and it outperforms model-tuning physical parameters for CD prediction. This enables non-destructive scatterome- try to detect a CD drift of 1nm during a multi-electron beam lithography process.
- In publication [47], a DNN with two fully connected hidden layers is employed to link Imaging Spectroscopic Reflectometry (ISR) to Bottom CD (BCD). The study references the BCD from SEM images of high aspect ratio channel holes. The DNN model achieves a R2 value of 0.96. The study also links the predicted BCD value with defectivity, paving the way for future work on feeding back and adjusting the etching recipe in a timely manner.

The above examples show that the NN model can extract features from spectra data better than other ML methods. In the study by M. Terzi et al. [43], a more complicated type of data, time-series spectra, is addressed with a deep learning CNN model. Additionally, in the study by N. Figueiro et al. [31], when compared with the traditional spectra model fitted by a physical model, the NN model provides better accuracy.

On this matter, it is also important to notice that in semi- conductor manufacturing, the raw collected data is often com- plex, multimodal, and multicollinear. Therefore, data prepro- cessing,

mining, and cleaning are crucial steps for achieving good predictions. In fact, many publications deal with large amounts of data with complex relations between the features, necessitating the use of dimensional reduction methods to extract meaningful information and visualize complex data. For example, the publication of C. H. Chen et al. [20] analyzes a thin-film process step, Physical Vapor Deposition (PVD), using a virtual metrology method based on a combination of tree-based models. For the data preparation step, the authors have decided to reduce the dataset to two dimensions (2D) using a nonlinear dimensionality reduction technique called t- distributed Stochastic Neighbor Embedding (t-SNE) to analyze outliers. This work explores how new dimensionality reduction techniques, capable of addressing non-linear datasets, can be extremely useful in semiconductor manufacturing optimization.

VI. CHALLENGES AND LIMITATIONS OF CURRENT ML TECHNOLOGIES

Integrating machine learning (ML) into semiconductor manufacturing domain poses a dual set of hurdles, encompassing both technical and practical obstacles, as well as inherent limitations within existing ML technologies.

A. Challenges

1) *Scalability and Computational Resources:* The primary bottleneck in training ML models at an industrial level lies in the requirement for extensive semiconductor manufacturing datasets, necessitating significant computational resources and infrastructure. Conversely, scalability becomes a major concern when deploying ML solutions across various manufacturing facilities or integrating them into live production environments. While Federated Machine Learning (FedML) can be helpful in this context, it also presents challenges such as ensuring data security and privacy, as well as effectively synchronizing and aggregating model updates.

2) *Robustness and Generalization:* Machine learning models trained on datasets from previous process conditions may face challenges when generalizing to new process conditions or unforeseen events, especially in anomaly prediction scenarios. Ensuring the robustness and adaptability of these models to evolving process dynamics is imperative for maintaining their long-term effectiveness.

3) *Data Quality and Quantity*: Although FABS

generate vast amounts of data, the data pertinent to ML applications often suffer from noise, bias, or incompleteness, resulting in suboptimal model performance. Furthermore, obtaining labeled data, particularly for supervised model training, can be both costly and time-consuming.

4) *Feature Engineering:* Recognizing relevant features and extracting valuable insights from raw data are pivotal for optimizing ML model performance. Nonetheless, conducting feature engineering in semiconductor manufacturing poses challenges due to the multidimensional nature of the data (as well as due to highly complex and variable Semiconduc- tor manufacturing processes) and the necessity to capture subtle process variations (while considering equipment vari- ations, material properties, and environmental factors simulta- neously).

5) Interpretability and Explainability: ML models employed in semiconductor manufacturing often functions as black-boxes, presenting challenges in comprehending their decisions or providing explanations for model predictions.. This lack of interpretability thwarts confidence and acceptance, particularly in safety-critical applications like automotive or aerospace, necessitating stringent regulatory and compliance standards.

6) Domain Expertise and Knowledge Fusion: There is a crucial need for the best of both worlds: on one side, expertise in ML algorithms is essential, while on the other, semiconductor process domain knowledge is highly significant for the development and deployment of effective solutions in semiconductor manufacturing. Nevertheless, a scarcity of proficient individuals possessing expertise in both semiconduc- tor engineering and ML exists, underscoring the necessity for interdisciplinary collaboration.

B. Limitations

1) ML Model Limitations: Capturing the intricate non- linear relationships inherent in semiconductor manufactur- ing data poses a challenge for any single or specific ML model/technique. Additionally, generalization remains a per- sistent issue across various process steps. Advanced DL techniques or Multi-modal ML approaches could potentially offer improved performance by analyzing large datasets and establishing connections between different process steps, such as Litho to Etch to CMP, enhancing the overall understanding and efficiency of the manufacturing process.

2) *Underfitting, Overfitting with Data Imbalance:* In semi- conductor manufacturing processes, our primary focus often lies on stochastic failures or events occurring within tools or

processes. However, relevant data tends to be insufficient or imbalanced, which can result in the model being underfitted or overfitted, ultimately leading to poor generalization perfor- mance. Ensuring model reliability remains a concern unless we address the issue of data imbalance and ensure adequate representation of rare events.

3) *Ethical and Bias Concerns*: ML models trained on biased semiconductor manufacturing process data may sig- nificantly impact the accuracy of process control. Ensuring fairness in ML applications necessitates thorough attention to bias mitigation techniques and adherence to ethical guidelines. *4*) *Privacy and Security Risks*: Semiconductor manufactur- ing data often contains sensitive information about proprietary processes, materials, or designs. Protecting data privacy and security is essential to prevent unauthorized access, data breaches, or intellectual property theft.

lenges and limitations of recent ML applications, aiming to address them in the near future through collaboration among semiconductor engineers, data scientists, ethicists, regulatory experts, and other stakeholders. By leveraging the capabilities of advanced ML technologies and addressing ethical concerns, we can explore new horizons in advanced process control at advanced IC nodes.

In this section, we have addressed various practical chal-

VII. OUTLOOK

ML methods not only successfully optimize semiconductor processes but also save more computational power com- pared with conventional physics models that solve Maxwell's equations [33]. Even on a molecular scale [51], data-driven ANN yield relatively good results when validated with Com- putational Fluid Dynamics (CFD) models. In scatterometry metrology, NN methods found a more accurate solution for critical dimension (CD) and film thickness than the traditional physics model, Rigorous Coupled Wave Analysis (RCWA) [76]. With deep learning CNN, even time series spectra [85] can be processed without any feature extraction algorithm, and the predictions showed accurate results. But usually, the ML models are trained by feeding output results from computational models, which are costly and rely on experience for tuning boundary conditions and discretizing geometry. Most ML-based models rely on model-based OPC. Therefore, a

model assumptions. Moreover, incorporating data from various modalities, such as sensor data, imaging data, and process parameters, can offer a more comprehensive understanding of semiconductor manufacturing processes. On the other hand, enhancing the interpretability and explainability of multi- modal ML models is essential for gaining insights into model predictions and building trust among stakeholders. Techniques such as attention mechanisms, model visualization, and expla- nation generation can facilitate understanding the underlying factors influencing model decisions.

VIII. CONCLUSIONS

In this systematic review, 58 papers, since 2015, using machine learning to facilitate semiconductor processing are analyzed. We briefly addressed and discussed the principal process concepts and key challenges at each process step, followed by an exhaustive study of various machine learning methods, their application purposes, validation results, and limitations. Our analysis of these publications clearly shows that the most prevalent application processes are lithography and metrology. In recent years, there has been a growing trend in the implementation of neural networks, due to their unparalleled capability for handling non-linear, complex data with higher accuracy compared to other ML methods. In some data ML method that could integrate physical law, symmetry and conservation law will help data-driven ML model performs better, especially for some applications data is expensive or limited. In recent years, physics-informed machine learning [86] has shown great potential in overcoming the low data availability problem and solving ill-posed or inverse problems compared to traditional physical models (CFD, RWCA, CTM ...etc). Some physics-assisted research for semiconductor pro- cesses [71] already shows positive results when physics-based training results are transferred to neural network models. In the research, both transfer learning (TL) and semi-supervised multi-task learning (SS-MTL) outperform the baseline Multi- Layer Perceptron (MLP), which is a NN model without any knowledge of physics equations. More research on physics- informed machine learning in semiconductor manufacturing should be reviewed and analyzed. The integration of data- driven methods and knowledge of physics laws will be in- dispensable and ubiquitous in the 4th industrial revolution, especially in the complicated but fastidious semiconductor industry. Additionally, it is crucial to develop techniques for quantifying uncertainty in ML models to assess model reliability and make informed decisions in semiconductor manufacturing. Probabilistic modeling approaches can effec- tively capture uncertainty stemming from data variability and

structure inputs, i.e., spectra (ellipsometry or OES), almost all the studies adopted neural networks regardless of their related processes or application purposes. Nevertheless, different ML methods have been proven capable of increas- ing the throughput of various processes and purposes in semiconductor manufacturing, helping to optimize or control the process more easily. Many predictive metrology stud- ies have proven the potential to discover process problems months/weeks before the final test or probe some proper- ties in a non-destructive manner, compared to their previous counterparts. The application of machine learning techniques in semiconductor process optimization is still in its very early stages. Numerous challenges remain to be addressed, presenting significant opportunities for future research. Future research may explore: (a) discovering and characterizing novel materials by analyzing existing and in-use vast datasets of material properties and performance metrics., (b) creating detailed simulations (digital twins) of semiconductor processes to allow virtual validation of new process recipes and materi- als, reducing the cost and time associated with experimental processes, (c) enabling real-time process control to contin- uously monitor and adjust parameters, maintaining optimal conditions and enhancing semiconductor yield, (d) implement- ing collaborative learning systems by integrating Mixture-of- Experts (multi-modal machine learning models/algorithms) to share insights and improve overall process optimization in large-scale semiconductor manufacturing environments with multiple production lines (for IDMs, foundries, fabs), finally (e) focusing on optimizing resource usage (such as water and energy), reducing waste, and improving recycling processes to make semiconductor manufacturing more sustainable. Ex- ploring these potential directions and integrating advanced ma- chine learning paradigms like Graph Neural Networks (GNN), Physics-Informed Neural Networks (PINN), and Generative AI (GenAI) into semiconductor process optimization hold the potential to revolutionize the industry, driving it towards greater efficiency, innovation, and sustainability.

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